

Question	Marking Guidance	Mark	Comments
02.1	1. Large(r) organisms have a small(er) surface area:volume (ratio); OR Small(er) organisms have a large(r) surface area:volume (ratio); 2. Overcomes long <u>diffusion</u> pathway OR Faster <u>diffusion</u>;	2	2. Accept short diffusion pathway 2. Accept for 'faster', more

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02.2	Mark in pairs, 1, and 2 OR 3. and 4. 1. Water has low(er) oxygen partial pressure/concentration (than air); 2. So (system on outside) gives large surface area (in contact with water) OR So (system on outside) reduces diffusion distance (between water and blood); 3. Water is dense(r) (than air); 4. (So) water supports the systems/gills;	2	

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02.3	1. In fish, blood leaving (V) has more oxygen than water leaving (E); 2. (But) in humans, blood leaving (V) has less oxygen than air leaving (E); 3. Difference in oxygen (concentration) between artery and vein is greater in fish than in humans; 4. (So) fish remove a greater proportion from the oxygen they take in;	2 max	

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02.4	1. Blood and water flow in opposite directions; 2. Diffusion/concentration gradient (maintained) along (length of) lamella/filament;	2	Accept for 2 marks, suitably labelled diagram

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02.5	1. and 2. Correct answer for 2 marks, 4.3 (times greater);; Accept for 1 mark, 4.333333333 (correct answer not given to 2 significant figures) OR Evidence of 130 (cm ³ kg ⁻¹) and 30 (cm ³ kg ⁻¹) Correct explanation for 1 mark, 3. Provides more oxygen for <u>respiration</u> ;	3	

